

RESECTION RACCOON - TEACHER KEY

Use the teacher answer map. Accept a small error area for pencil width, printing, and landmark-center choices.

Answer summary

START: Langley 331 -> draw 151; Brashear 125 -> draw 305.

RR1: Prospect 111 -> draw 291; St. Margaret 231 -> draw 051.

RR2: St. Margaret WSW/247.5 -> draw ENE/067.5; Brashear 110 -> draw 290.

RR3: Brashear NE/045 -> draw SW/225; St. Martin SSE/157.5 -> draw NNW/337.5.

Challenge notes

- 1 - Student team** Answer: START marker in the West End/Westwood portion of the map.
- 2 - Raccoon 1** Answer: RR1 marker. Require both schools in the radio report.
- 3 - Raccoon 2** Answer: RR2 marker. Students translate WSW, then reverse both sightings.
- 4 - Raccoon 3** Answer: RR3 marker. Students translate and reverse NE and SSE.

Suggested ZULU MAIN response

"[TEAM], this is ZULU MAIN. Copy your fix for [CALL SIGN]. Confirm the two landmarks and repeat the location. Over."

If wrong, ask only: "Did you draw the measured azimuth or the back azimuth from the school?"

Assessment checks

- Identifies each landmark before requesting a bearing.
- Requests at least two useful, nonparallel sightings.
- Converts direction language to azimuths.
- Reverses reported sightings before drawing from schools.
- Reports and defends the intersection clearly to ZULU MAIN.

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Do not volunteer all four sightings. Give one response per clear radio question. Two strong lines are enough; allow a third sighting if students choose a weak crossing angle or want to confirm their fix.

RESECTION RACCOON 4 - NUMERICAL BEARINGS

Answer location: RR4 marker

Opening: "ZULU MAIN, this is RESECTION RACCOON 4. I am lost. I have a compass" and can identify several schools. Over."

School RR4 reports Teacher back azimuth

Langley High School	325	145
St. Philip School	277	097
Prospect School	095	275
Brashear High School	127	307

Strong pairs: Langley + Prospect; St. Philip + Brashear; or Langley + Brashear.

RESECTION RACCOON 5 - DIRECTION LANGUAGE

Answer location: RR5 marker

Opening: "ZULU MAIN, this is RESECTION RACCOON 5. I am lost. I can report" compass directions to four schools. Over."

School RR5 reports Draw from school

St. Cyril School	NW / 315	SE / 135
Carmalt School	W / 270	E / 090
Prospect School	ENE / 067.5	WSW / 247.5
Pioneer Education Center	SE / 135	NW / 315

Strong pairs: Carmalt + Prospect; St. Cyril + Pioneer; or St. Cyril + Prospect.